

History of Hanoi

- * The tower of Hanoi is a famous mathematical puzzle invented in 1883 by the French mathematician Edouard Lucas.

History of the Tower of Hanoi

- * Edouard Lucas introduced the puzzle in 1883.
- * The puzzle consists of three rods (towers) and a number of disks of different size stacked on one rod, with largest disk at the bottom and the smallest at the top.
- * The objective is to move all the disks from the source rod to the destination rod while following these rules.

- a. Only one disk can be moved at a time
- b. Only the top disk of a rod can be moved.
- c. A larger disk cannot be placed on a smaller disk.

The legend.

Lucas popularized the puzzle with a legend about a temple in Hanoi, Vietnam. According to the story.

- * In a sacred temple, priests were moving 64 golden disks among three diamond needles.
- * They followed the same rules as the

Tower of Hanoi puzzle

- * The legend says that when the priests complete the task, the world will end.

Mathematical Importance

The tower of Hanoi is widely used in computer science to teach.

* Recursion

* Algorithm

* problem-solving techniques

The minimum number of moves required to solve the puzzle with n disks is.

$$2^n - 1$$

For example:

- 3 disk $\rightarrow$ 7 moves
- 4 disk $\rightarrow$ 15 moves
- 5 disk $\rightarrow$ 31 moves.

For the legendary 64 disks, it would take:

$$2^{64} - 1$$

= 18,446,744,073,551,615 moves; an

enormous number that would take billions of years to complete.

Thus, the Tower of Hanoi remains on the most famous puzzles in mathematics and computer science.